

NodeMedic-FINE: Automatic Detection and Exploit Synthesis for Node.js Vulnerabilities

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The Node.js ecosystem comprises millions of packages written in JavaScript. Many packages suffer from vulnerabilities such as arbitrary code execution (ACE) and arbitrary command injection (ACI). Prior work has developed automated tools based on dynamic taint tracking to detect potential vulnerabilities, and to synthesize proof-of-concept exploits that confirm them, with limited success.

One challenge these tools face is that expected inputs to package APIs often have varied types and object structure. Failure to call these APIs with inputs of the correct type and with specific fields leads to unsuccessful exploit generation and missed vulnerabilities. Generating inputs that can successfully deliver the desired exploit payload despite manipulation performed by the package is also difficult.

To address these challenges, we use a type and object-structure aware fuzzer to generate inputs to explore more execution paths during dynamic taint analysis. We leverage information generated by the taint analysis to infer the types and structure of the inputs, which are then used by the exploit synthesis engine to guide exploit generation.

We implement NodeMedic-FINE and evaluate it on 33,011 npm packages that contain calls to ACE and ACI sinks. Our tool finds 2257 potential flows and automatically synthesizes working exploits in 766 packages.

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